Farming in Space
The VEGGIE experiment on board the ISS is used for harvesting crops, and serves as a test for future deep space missions. **https://dgit.in/may19-73**

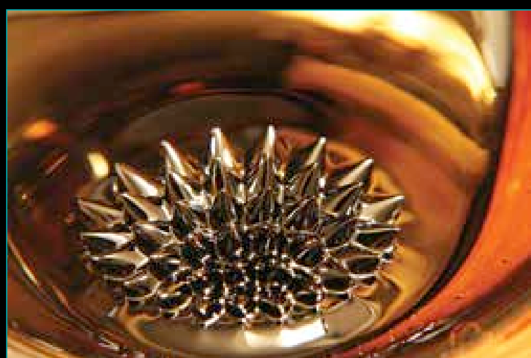
Returning to Earth
How difficult really is it to walk on the surface of the Earth after spending 6 months in space? **https://dgit.in/may19-74**

more closely packed, allowing for both a hard and reflective gold coating. The same process is also used to coat the gold on the Oscars, after previous winners started complaining that the gold coating was flaking off.

Phase changing materials or PCMs were originally developed by NASA for the bodysuits of astronauts for temperature regulation. If conditions got too warm, the material would store the heat and prevent it from reaching the body. If the conditions got too cold, the same material would release the stored heat. In most cases, it was used for keeping the astronauts cooler. NASA developed a number of PCMs for different target temperatures. These materials were transformed into a paint form, and used to keep the Shuttles cooler. Later the technology was also used in aeroplanes and “iceless” coolers that could store cold food or beverages for up to 2 to 4 hours longer than regular containers. The Jockey Staycool range of innerwear uses the same technology to provide temperature regulation to its consumers. At the other end of the temperature gradient, NASA’s flexible aerogel insulation which was used in the tanks storing the fuel for the space shuttle, is also used by a company called Oros to make extremely lightweight outerwear for cold conditions – such as trekking in high altitude mountain ranges.

SUSTAINABILITY

The many Earth imaging satellites by the space agency are used in various ways to reduce damage to the planet, including tracking forest fires, deforestation, changes in the atmosphere, global temperature levels and the extent of ice caps. NASA deals with small cocoons of human habitation in space, which are tiny ecosystems in themselves, such as the International Space Station. In such a closed environment, it makes sense to make the most out of the metabolic wastes of the human body. One of the early studies by NASA into reusing this waste was to employ hydrogen-fixing bacteria, to convert carbon dioxide into usable byproducts. Although the technology was itself not used by NASA in space, the study led to the technology being used on Earth. Earth can also face the same problem of a buildup of carbon dioxide, only on a much larger scale. The hydrogen fixing bacteria occupy very little space, and unlike other organisms can work in the dark, without the need for light. Kiverdi is a company that uses microorganisms to convert carbon dioxide into proteins and oils, which can then be used in a wide variety of industries, from providing low cost fish food, to ice cream and soap production. In the future, the same technology may even be used to make food for humans. Another form of organism, exoelectrogenic bacteria have the ability to transfer electrons outside their bodies. This property allows them to convert carbon dioxide into oxygen, water and methane. While the oxygen and water can be used in life support systems in a spacecraft, the methane can potentially be used as a source for fuel. This technology is used in the EcoVolt reactor is a wastewater treatment solution, that produces clean water, and high quality biogas as byproducts. The biogas is high in methane content, and can be used to produce clean energy.



Ferrofluid spikes

HEALTHCARE

Polytetrafluoroethylene (PTFE) better known as Teflon, and is used for nonstick coating on pans. The variant known as ePTFE is a thinner, stronger and stiffer variation that was developed for the exacting conditions required to collect samples on the MARS rovers. The requirements were that the material could be sterilized and did not react to the environment. The properties of the material turned out to be useful for saving human lives as well. The material is now used for making stents for heart patients, and for sutures that can be harmlessly left within the human body, without the need to be removed.

For years, NASA Doctors at Johnson Space Center had been studying damage to the DNA of cells of astronauts by studying samples of blood before and after spaceflight, because of the high amount of radiation they were exposed to. The damage could include translocations (the DNA strand is not where it should be), inversions (portions of the DNA are inverted) or other kind of DNA variations. If the cells die out, the mutations are not a problem. However, if the cells can replicate, it can result in cancerous growth. To better study the cellular damage, NASA doctors funded the development of an approach that would help easily identify the variations. The approach tags DNA strands frozen in the middle of replication with synthetic DNA, which serve as fluorescent chromatid paints. The technology developed had a number of medical uses. The approach could be used to reduce or increase the dosage of therapies for cancer patients, depending on the amount of cell damage observed. It could be used to identify cancer causing mutations, or mutations related to other diseases. The technology could also be used to identify ideal candidates for clinical trials, and has potential future applications in gene editing.